

Logical to Physical Address Conversion Examples

Example Problem 1

Problem Statement

A paging system has:

- Logical address space: 16 bits
- Physical address space: 14 bits
- Page size: 1 KB = 2^{10} bytes

Given:

- Logical address: 0x1A5C
- Page table mapping: Page 0x6 → Frame 0x2A

Step 1: Determine Page Number and Offset

Since page size = 2^{10} , the offset requires 10 bits.

Offset bits = 10

Logical address = 16 bits

Page number bits = $16 - 10 = 6$

0x1A5C = 0001 1010 0101 1100■

- Page number = 000110■ = 0x6
- Offset = 1001011100■ = 0x25C

Step 2: Page Table Lookup

Page 0x6 → Frame 0x2A

Step 3: Construct Physical Address

Physical address format:

Frame Number || Offset

Physical Address = 0xAA5C

Final Answer

Physical Address = 0xAA5C

Example Problem 2

Problem Statement

A paging system has:

- Logical address space: 20 bits
- Physical address space: 18 bits
- Page size: 4 KB = 2^{12} bytes

Given:

- Logical address: 0x3F2B9
- Page table mapping: Page 0x3F → Frame 0x1C

Step 1: Determine Page Number and Offset

Page size = 2^{12}

Offset bits = 12

Page number bits = $20 - 12 = 8$

0x3F2B9 = 0011 1111 0010 1011 1001■

- Page number = 00111111■ = 0x3F
- Offset = 0010 1011 1001■ = 0x2B9

Step 2: Page Table Lookup

Page 0x3F → Frame 0x1C

Step 3: Construct Physical Address

Frame = 011100■

Offset = 0010 1011 1001■

Physical Address = 011100001010111001■

Hexadecimal = 0x1C2B9

Final Answer

Physical Address = 0x1C2B9

Summary Table

Parameter	Problem 1	Problem 2
Logical Address Bits	16	20
Physical Address Bits	14	18
Page Size	1 KB	4 KB
Offset Bits	10	12
Page Bits	6	8
Frame Bits	4	6
Final Physical Address	0xAA5C	0x1C2B9